

Yangzong Baima

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Profile:

MSc HCI candidate (UCD) with a BEng in Civil Engineering (Tongji) — bridging structural domain knowledge with AI and user-centred design. Hands-on experience across geotechnical analysis, heritage condition assessment, and LLM data pipelines. Targeting ConTech roles where construction expertise meets digital innovation.

Education:

University College Dublin (UCD)	2025.08–2026.09
MSc in Human–Computer Interaction	Dublin, Ireland
Tongji University	2018.08 – 2023.06
Bachelor of Civil Engineering	Shanghai, China

Work Experience:

Tibet Autonomous Region Institute of Architectural Survey & Design Heritage Conservation Intern	2020.07 Lhasa, China
- Diagnosed differential settlement and roof cracking at Baima Monastery (Shannan) through on-site structural inspection, identifying cumulative waterproofing overload as the root cause. - Produced AutoCAD remediation drawings of the affected residential compound, translating field measurements into construction-ready documentation for the repair team. - Contributed to a condition assessment report under national heritage standards, applying regional construction knowledge to propose preservation-compliant intervention strategies.	
Tibet Xingguang Social Work Service Center Data Processor	2024.11–2025.05 Lhasa, China
- Engineered a custom VBA automation tool that reduced manual data entry time by 40% and standardized file management protocols for the entire organization. - Managed demographic datasets for over 100 service recipients using Stata, ensuring 100% accuracy in government reporting and service tracking. - Orchestrated over 100 registration submissions for the regional Volunteer Project Competition, handling sensitive user data with zero errors. - Developed comprehensive documentation templates for service follow-ups, establishing a standardized digital filing system.	
Wiseweb Technology Group Co., Ltd. Corpus Engineer	2024.05–2024.10 Lhasa, China
- Curated and cleaned over 4,000 specific linguistic data entries in medical and financial fields using Python to support Large Language Model (LLM) training. - Streamlined data collection workflows and optimized JSON/CSV storage formats, improving accessibility for cross-functional technical teams. - Collaborated with engineering teams to synchronize audio and text data, ensuring high-quality inputs for speech recognition model alignment. - Implemented rigorous data cleaning protocols using fast_align to create high-precision translation pairs.	

Project Experience:

Museum Deep Foundation Pit Retaining Design (Tongji Capstone):

2023.02–06

- **Designed** a full retaining system for a 6.5m-deep museum excavation (Hangzhou, 125,200m² site): selected bored pile + twin-head CSM watercut curtain over alternatives through multi-criteria scheme comparison per national and Zhejiang provincial standards.
- **Computed** pile internal forces, deformation envelopes, and stability safety factors (global sliding $K=3.56$, base heave $K=2.2$) across three excavation stages using equivalent-stiffness method and Tongji parabolic surface-settlement model.
- **Delivered** 6 AutoCAD construction drawings (site plan, retaining layout, strut plan, cross-section, details) and a 15,000-word bilingual design report with full monitoring scheme.

Central Statistics Office (CSO) Visualization Project:

2026.01

- Transformed raw census data into an interactive dashboard using Vega-Lite to interpret complex demographic trends for government stakeholders.
- Delivered an on-site demonstration of the dynamic dashboard's interactive features to over 20 staff members at the CSO headquarters.

Interests & Skills:

- **Technical Skills:** Python, Excel, R, Vega-Lite, SQL, Stata, C++, VBA, AutoCAD.
- **AI & Development:** AI-Augmented Coding (Prompt Engineering, Copilot), Rapid Prototyping.
- **Design & Visualization:** Figma, Tableau, Power BI, Microsoft Office.
- **Languages:** English (Professional), Chinese (Native), Tibetan (Native).
- **Interests:** Data Storytelling, ConTech & Digital Construction.